

# DNS Threat Hunting and Suspicious Domain Analysis

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## Executive Summary

Domain Name System (DNS) traffic provides valuable visibility into communication between enterprise endpoints and external infrastructure. Because almost every network connection begins with a DNS lookup, DNS telemetry is frequently used by Security Operations Centers (SOCs) to detect malware, command-and-control (C2) communications, phishing campaigns, and Domain Generation Algorithm (DGA) activity.

This case study demonstrates a DNS threat hunting workflow implemented in Python. DNS logs were enriched with domain frequency scores and compared against the Cisco Umbrella Top 1 Million domain list to identify uncommon domains, naked IP lookups, and suspicious DNS activity. The investigation illustrates how multiple analytical techniques can be combined to prioritize domains requiring further investigation.

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## Objective

The objective of this investigation was to analyze enterprise DNS logs and enrich the collected data using multiple analytical techniques in order to identify potentially suspicious DNS activity.

The investigation specifically aimed to:

- Analyze DNS query logs.
  - Extract root domains from DNS requests.
  - Calculate statistical frequency scores for each domain.
  - Compare queried domains against the Top 1 Million domain list.
  - Detect naked IP address lookups.
  - Evaluate suspicious domains using probability-based frequency scoring.
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## Investigation Environment

### Platform

- Python
- Jupyter Notebook

### **Libraries**

- Pandas
- Regular Expressions (re)
- FreqCounter

### **Data Sources**

- Enterprise DNS Logs
  - Frequency Table (2018)
  - Top 1 Million Domains Dataset
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## **Investigation Methodology**

The investigation consisted of several sequential enrichment stages.

### **Stage 1 – DNS Log Collection**

The DNS log dataset was imported into Python for analysis.

A total of:

**29,368 DNS records**

were processed during the investigation.

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### **Stage 2 – Root Domain Extraction**

Each DNS query was normalized by extracting the root domain while removing unnecessary prefixes.

This preprocessing step ensured that subsequent statistical analysis was performed against standardized domain names rather than raw DNS queries.

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### **Stage 3 – Domain Frequency Analysis**

Each extracted domain was evaluated using a statistical frequency model.

Domains with lower probability scores are generally less common and may indicate:

- Domain Generation Algorithms (DGAs)
  - Newly registered domains
  - Randomly generated domains
  - Potential attacker-controlled infrastructure
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#### **Stage 4 – Top 1 Million Domain Validation**

Each domain was compared against a reference dataset containing:

##### **891,544 unique domains**

Domains present within the Top 1 Million list were classified as common domains, while domains outside the reference dataset received additional attention during the investigation.

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#### **Stage 5 – DNS Log Enrichment**

The original DNS logs were enriched with additional contextual information including:

- Root domain
- Frequency score
- Top 1 Million status
- Source IP address

This enrichment process transformed raw DNS logs into actionable investigation data.

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#### **Stage 6 – Naked IP Detection**

Several IP addresses were evaluated to determine whether they resolved to known domains.

Results included:

| IP Address | Result |
|------------|--------|
|------------|--------|

|                |          |
|----------------|----------|
| 137.245.120.50 | Naked IP |
|----------------|----------|

|              |                                                      |
|--------------|------------------------------------------------------|
| 104.244.42.1 | <a href="http://www.twitter.com">www.twitter.com</a> |
|--------------|------------------------------------------------------|

|              |          |
|--------------|----------|
| 172.20.56.90 | Naked IP |
|--------------|----------|

|              |          |
|--------------|----------|
| 99.84.181.52 | Naked IP |
|--------------|----------|

Naked IP communication can be indicative of malware, command-and-control infrastructure, or manually configured network connections and therefore warrants additional investigation.

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## Stage 7 – Suspicious Domain Evaluation

Several domains were evaluated using statistical frequency scoring.

| Domain | Frequency Score |
|--------|-----------------|
|--------|-----------------|

|       |        |
|-------|--------|
| zak0n | 2.0316 |
|-------|--------|

|        |        |
|--------|--------|
| mitula | 8.3011 |
|--------|--------|

|          |        |
|----------|--------|
| ae0333a7 | 1.4058 |
|----------|--------|

|         |        |
|---------|--------|
| M0M3367 | 1.7917 |
|---------|--------|

Domains exhibiting unusually low frequency scores may represent randomly generated or attacker-controlled domains and should be prioritized for additional investigation using threat intelligence or passive DNS data.

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## Key Findings

The investigation successfully:

- Processed **29,368 DNS log records**.
- Extracted and normalized root domains.
- Enriched DNS logs with statistical frequency scores.
- Compared domains against **891,544** known domains.

- Identified multiple naked IP communications.
  - Evaluated suspicious domains using frequency analysis.
  - Generated an enriched DNS dataset suitable for further security analytics.
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### **Skills Demonstrated**

- DNS Threat Hunting
  - Security Analytics
  - DNS Log Enrichment
  - Domain Frequency Analysis
  - Threat Intelligence Preparation
  - Python Programming
  - Data Analysis
  - Threat Hunting Methodology
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### **Security Impact**

DNS traffic often provides one of the earliest indicators of malicious activity within enterprise networks.

The workflow demonstrated in this project enables analysts to prioritize DNS events by combining:

- Statistical domain analysis
- Reputation checking
- IP-to-domain mapping
- Domain frequency scoring

These techniques are commonly employed within Security Operations Centers to detect malware communications, phishing campaigns, command-and-control infrastructure, and anomalous DNS activity.

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## **Conclusion**

This investigation demonstrates a practical DNS threat hunting workflow that transforms raw DNS logs into enriched investigative data. By combining domain normalization, statistical frequency scoring, Top 1 Million validation, and naked IP detection, the investigation successfully prioritizes suspicious DNS activity for further analysis.

The methodology illustrated in this case study reflects real-world enterprise security analytics techniques and provides a scalable approach for identifying potentially malicious DNS activity within large datasets.